HW00

Course: Description

January 01, 2026 at 23:59:00

Instructor

Vai Srivastava

Problem 1

Parts *(# pts)*
a) Instructions *(# pts)*
b) Instructions *(# pts)*

a) Answer

b) Answer

Problem 2

Code *(# pts)*

Instructions

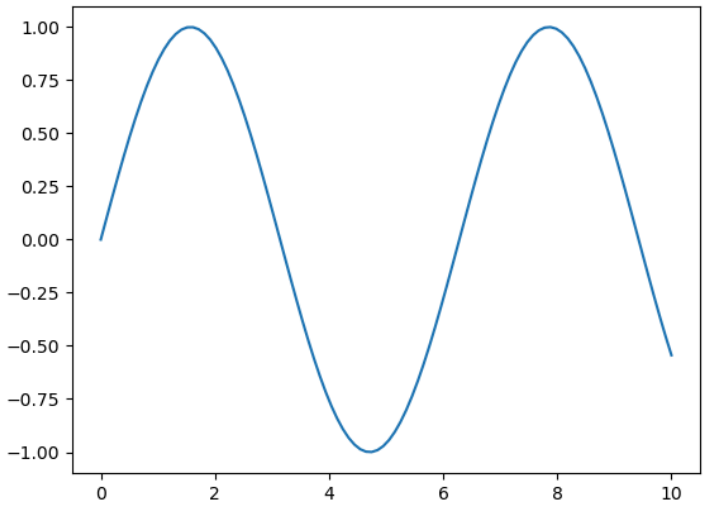


Figure 1: Sine Wave on $x = [010]$

```
x:
[ 0.          0.1010101   0.2020202   0.3030303   0.4040404   0.50505051
  0.60606061  0.70707071  0.80808081  0.90909091  1.01010101  1.11111111
  1.21212121  1.31313131  1.41414141  1.51515152  1.61616162  1.71717172
  1.81818182  1.91919192  2.02020202  2.12121212  2.22222222  2.32323232
  2.42424242  2.52525253  2.62626263  2.72727273  2.82828283  2.92929293
  3.03030303  3.13131313  3.23232323  3.33333333  3.43434343  3.53535354
  3.63636364  3.73737374  3.83838384  3.93939394  4.04040404  4.14141414
  4.24242424  4.34343434  4.44444444  4.54545455  4.64646465  4.74747475
  4.84848485  4.94949495  5.05050505  5.15151515  5.25252525  5.35353535
  5.45454545  5.55555556  5.65656566  5.75757576  5.85858586  5.95959596
  6.06060606  6.16161616  6.26262626  6.36363636  6.46464646  6.56565657
  6.66666667  6.76767677  6.86868687  6.96969697  7.07070707  7.17171717
  7.27272727  7.37373737  7.47474747  7.57575758  7.67676768  7.77777778
  7.87878788  7.97979798  8.08080808  8.18181818  8.28282828  8.38383838
  8.48484848  8.58585859  8.68686869  8.78787879  8.88888889  8.98989899
  9.09090909  9.19191919  9.29292929  9.39393939  9.49494949  9.5959596
  9.6969697   9.7979798   9.8989899  10.        ]

y:
[ 0.          0.10083842  0.20064886  0.2984138   0.39313661  0.48385164
  0.56963411  0.64960951  0.72296256  0.78894546  0.84688556  0.8961922
  0.93636273  0.96698762  0.98775469  0.99845223  0.99897117  0.98930624
```

```
24   0.96955595  0.93992165  0.90070545  0.85230712  0.79522006  0.73002623
25   0.65739025  0.57805259  0.49282204  0.40256749  0.30820902  0.21070855
26   0.11106004  0.01027934 -0.09060615 -0.19056796 -0.28858706 -0.38366419
27  -0.47483011 -0.56115544 -0.64176014 -0.7158225  -0.7825875  -0.84137452
28  -0.89158426 -0.93270486 -0.96431712 -0.98609877 -0.99782778 -0.99938456
29  -0.99075324 -0.97202182 -0.94338126 -0.90512352 -0.85763861 -0.80141062
30  -0.73701276 -0.66510151 -0.58640998 -0.50174037 -0.41195583 -0.31797166
31  -0.22074597 -0.12126992 -0.0205576   0.0803643   0.18046693  0.27872982
32   0.37415123  0.46575841  0.55261747  0.63384295  0.7086068   0.77614685
33   0.83577457  0.8868821   0.92894843  0.96154471  0.98433866  0.99709789
34   0.99969234  0.99209556  0.97438499  0.94674118  0.90944594  0.86287948
35   0.8075165   0.74392141  0.6727425   0.59470541  0.51060568  0.42130064
36   0.32770071  0.23076008  0.13146699  0.03083368 -0.07011396 -0.17034683
37  -0.26884313 -0.36459873 -0.45663749 -0.54402111]
```

See the Python code for this problem.

Code

./src/index.py Python

```python
import os

import matplotlib.pyplot as plt
import numpy as np


def heading(problem: str) -> None:
    print(f"\n------\n {problem}\n------")


def main() -> None:
    #########
    # p02 #
    #########
    heading("p02")

    x = np.linspace(0, 10, 100)
    y = np.sin(x)

    print(f"x:\n{x}\n\ny:\n{y}")
    with open("./outputs/text/s02.txt", "w", encoding="utf-8") as f:
        f.write(f"x:\n{x}\n\ny:\n{y}")

    plt.figure()
    plt.plot(x, y)

    save_path = os.path.join(os.path.dirname(__file__), "./outputs/figures/s02.png")
    plt.savefig(save_path)
    plt.show()


if __name__ == "__main__":
    main()
```